Figure 3.1 identifies the links, joints, and four actuator channels used throughout the model.

Figure 3.2 separates the rigid bodies that form the assembled mechanism. The catalog makes the L0–L5 body assignment explicit and shows the representative barrel–rod geometry used for cylinders C1–C4.

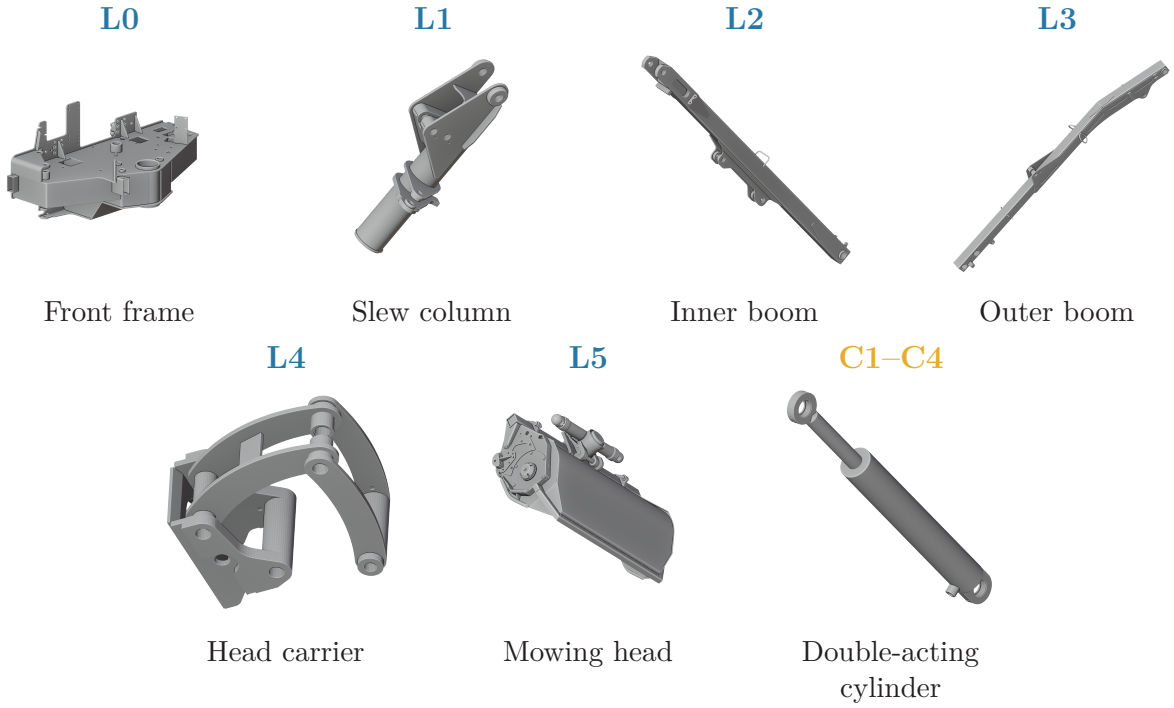


Figure 3.2: Individual CAD-derived bodies of the MULAG FME 500 shown at a common orthographic scale. L0–L5 are the front frame, slew column, inner boom, outer boom, head carrier, and mowing head. The C1–C4 panel shows the representative double-acting cylinder barrel–rod pair.

Manufacturer CAD supplies the link geometry, joint locations, masses, centers of mass and inertia tensors through Phobos [von Szadkowski 20]. The same CAD-derived bodies and joint frames form the mechanical basis of the Simscape and IsaacLab models keeping dimensions, axes, masses, and cylinder lever arms consistent while the hydraulic implementations remain model-specific.

3.2 Kinematic Structure and Coordinate Mappings

3.2.1 Joint Space and Cylinder Space

The configuration is $\boldsymbol{q} \in \mathbb{R}^5$; the actuated coordinates are the four cylinder extensions $\boldsymbol{x} \in \mathbb{R}^4$. Cylinder i joins CAD attachment points $\boldsymbol{p}_A$ and $\boldsymbol{p}_B$ on adjacent links. Its length and extension are

$$\ell_i(\boldsymbol{q}) = \left\| \boldsymbol{p}_B(\boldsymbol{q}) - \boldsymbol{p}_A(\boldsymbol{q}) \right\|, \qquad x_i = \ell_i(\boldsymbol{q}) - \ell_{0,i}, \tag{3.1}$$

with $\ell_{0,i}$ the retracted reference length. Differentiating (3.1) gives the scalar mapping between joint motion and extension rate,

$$\dot{x}_i = J_{\mathrm{cyl},i}(\boldsymbol{q})\, \dot{\boldsymbol{q}}, \qquad J_{\mathrm{cyl},i} = \boldsymbol{e}_i^{\top} \left(\boldsymbol{J}_B - \boldsymbol{J}_A \right), \tag{3.2}$$